WISER × Nestlé Quantum Challenge 2026

Technical Report

Adaptive Hybrid Quantum–Classical Optimization Framework for Distributed Order Management

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1. Business and Technical Summary

1.1 Introduction

Distributed Order Management (DOM) is the process of determining the most appropriate distribution center (DC) to fulfill customer orders within a supply chain network. When the default distribution center cannot completely satisfy an order due to constraints such as inventory shortages, throughput limitations, dock capacity, or operational restrictions, the order may be reassigned to an alternative distribution center. The objective is to maximize customer service and fulfillment value while minimizing shipping costs and penalties associated with unmet demand.

For global organizations such as Nestlé, efficient order fulfillment directly impacts customer satisfaction, operational efficiency, and logistics costs. As the number of customer orders and distribution centers increases, identifying the optimal assignment becomes increasingly difficult due to the large number of possible combinations. Therefore, advanced optimization techniques are required to support accurate and efficient decision-making.

1.2 Why Distributed Order Management is a Combinatorial Optimization Problem

Distributed Order Management is a combinatorial optimization problem because each customer order may have multiple feasible distribution centers, resulting in numerous possible assignment combinations. The number of feasible solutions increases rapidly with the number of orders, products, and distribution centers, making exhaustive evaluation computationally expensive.

The optimization process must simultaneously satisfy multiple operational constraints while maximizing overall business performance. These constraints include inventory availability, throughput capacity, dock capacity, shipping costs, and penalties for unmet demand. Since each order can be assigned to at most one distribution center, the problem requires selecting the optimal combination of assignments from a large search space.

1.3 Why Advanced Optimization Methods are Required

Traditional optimization approaches such as rule-based assignment or greedy heuristics provide fast decisions but often fail to achieve globally optimal solutions for large-scale logistics problems. Exact optimization methods can identify optimal solutions but become computationally expensive as the problem size increases.

Hybrid quantum-classical optimization provides an alternative approach by combining efficient classical preprocessing with quantum optimization techniques. Classical methods reduce the problem size and prepare the optimization model, while quantum algorithms explore the reduced solution space to identify high-quality assignment decisions. This hybrid strategy enables the evaluation of complex optimization problems within the limitations of current quantum computing technology.

1.4 Project Objectives

The primary objectives of this project are:

- To understand the Distributed Order Management problem and its business significance.
- To analyze the provided DOM datasets and identify feasible order-to-distribution-center assignments.
- To develop classical baseline solutions for comparison.
- To formulate the optimization problem as a Quadratic Unconstrained Binary Optimization (QUBO) model.
- To implement a hybrid quantum-classical optimization framework using QAOA.
- To apply a business repair strategy to ensure feasible assignment recommendations.
- To compare the performance of Greedy Baseline, Exact QUBO, and QAOA using common evaluation metrics.
- To analyze the scalability and practical limitations of the proposed framework.

1.5 Business and Technical Trade-offs

The proposed solution balances solution quality, computational efficiency, and business feasibility through a hybrid optimization framework.

Business Trade-offs

- Maximizing order fulfillment while minimizing shipping costs.
- Reducing penalties without violating operational constraints.

- Efficient utilization of inventory and distribution center resources.

Technical Trade-offs

- Greedy heuristics provide fast but locally optimal solutions.
- Exact optimization produces optimal solutions but has limited scalability.
- QAOA provides approximate solutions suitable for representative optimization problems within current quantum hardware limitations.
- Classical preprocessing and candidate reduction improve scalability by reducing the optimization search space before quantum optimization.

Overall Workflow of the Proposed Hybrid Quantum-Classical Framework

WISER DOM Dataset

↓

Data Preprocessing

↓

Feasible Candidate Generation

(352,702 → 133,495)

↓

Adaptive Candidate Reduction

(133,495 → 65,857)

↓

QUBO Formulation

↓

Hybrid Quantum-Classical Optimization

(Exact Solver + QAOA)

↓

Business Validation & Final Solution

The proposed framework begins with preprocessing of the WISER Distributed Order Management dataset, integrating order information with inventory availability, shipping cost, and operational capacity constraints. Feasible order-to-plant assignments are generated and filtered using business constraints. An adaptive priority scoring mechanism evaluates candidate assignments based on revenue, logistics cost, inventory availability, and operational limitations. The reduced candidate space is transformed into a Quadratic Unconstrained Binary Optimization (QUBO) model and solved using both classical exact optimization and Quantum Approximate Optimization Algorithm (QAOA). Finally, a business repair layer validates constraints and generates the final feasible order allocation.

2. Problem Formulation and Distributed Order Management Optimization Model

2.1 Distributed Order Management Problem Definition

Distributed Order Management (DOM) is a supply chain optimization problem that determines the most suitable distribution center (DC) for fulfilling customer orders while satisfying multiple operational constraints. The objective is to maximize business value by selecting assignments that provide high revenue contribution while minimizing logistics costs and avoiding capacity violations.

In the WISER DOM challenge, each customer order can potentially be fulfilled by multiple distribution centers. However, every assignment decision must satisfy constraints related to inventory availability, shipping distance, dock capacity, throughput limitations, and penalty conditions.

The optimization problem can be represented as selecting the best order-to-distribution-center assignment from a large set of feasible candidates.

2.2 Mathematical Representation

Let:

- $O = \{o_1, o_2, \ldots, o_n\}$represent the set of customer orders.
- $D = \{d_1, d_2, \ldots, d_m\}$represent available distribution centers.
- x_{ij}represent a binary decision variable:
- The optimization objective is to maximize the overall assignment score:
- $Maximize \sum_i \sum_j Score_{ij} x_{ij}$
- where $Score_{ij}$represents the adaptive priority value calculated from multiple business factors.

2.3 Objective Function

- The adaptive priority score combines important operational parameters:
- $Score_{ij} = w_1 R_{ij} - w_2 C_{ij} + w_3 I_{ij} + w_4 D_{ij} + w_5 T_{ij} - w_6 P_{ij}$
  where:
- R_{ij}= Revenue contribution
- C_{ij}= Shipping cost
- I_{ij}= Inventory availability
- D_{ij}= Dock capacity availability
- T_{ij}= Throughput availability
- P_{ij}= Penalty factor
- $w_1, w_2, \ldots, w_6$= optimization weights
  This formulation enables the model to balance profitability and operational feasibility.

2.4 DOM Constraints

Order Assignment Constraint

Each order must be assigned to only one distribution center:

$$\sum_j x_{ij} = 1$$

This prevents duplicate fulfillment decisions.

Inventory Constraint

The assigned distribution center must have sufficient inventory:

$$\sum_i Demand_i x_{ij} \leq Inventory_j$$

Throughput Constraint

The total assigned orders must not exceed DC processing capability:

$$\sum_i Volume_i x_{ij} \leq Throughput_j$$

Dock Capacity Constraint

Daily loading operations must remain within available dock capacity:

$$\sum_i x_{ij} \leq DockCapacity_j$$

2.5 Optimization Challenge

The original DOM dataset contains a very large search space because every order may have multiple possible fulfillment locations. Directly applying classical optimization methods becomes computationally expensive as the number of binary decision variables increases.

To overcome this scalability challenge, the proposed framework introduces:

- Feasible candidate generation
- Adaptive priority-based filtering
- QUBO transformation
- Hybrid quantum-classical optimization

This reduces the complexity while maintaining business feasibility.

3. Proposed Hybrid Quantum-Classical Optimization Framework

3.1 Framework Overview

The proposed framework combines classical optimization techniques with quantum computing methods to solve the Distributed Order Management (DOM) optimization problem efficiently. The framework follows a hybrid approach where classical preprocessing and problem reduction techniques are combined with quantum optimization methods to search for high-quality order assignment solutions.

The complete workflow consists of five major stages:

1. Data preprocessing and integration
2. Feasible candidate generation
3. Adaptive candidate reduction
4. QUBO-based quantum optimization
5. Business validation and solution evaluation

The framework reduces the original large-scale DOM problem into a smaller optimization problem suitable for quantum execution while preserving important business constraints.

3.2 Data Preprocessing and Integration

The first stage processes the raw WISER DOM dataset containing order information, distribution center details, inventory availability, shipping costs, and operational constraints.

The preprocessing stage performs:

- Data cleaning and missing value handling
- Integration of multiple data sources
- Generation of lookup tables for inventory, shipping cost, and capacity information

- Identification of possible order-to-distribution-center combinations

After preprocessing, the integrated dataset provides a complete representation of possible fulfillment decisions.

3.3 Feasible Candidate Generation

Each order may have multiple possible distribution centers. Therefore, the initial candidate generation stage creates all possible order-to-plant assignments and removes infeasible combinations.

The generated candidate space:

- Total possible assignments: **352,702**
- Feasible assignments after constraint checking: **133,495**

The feasibility filtering process considers:

- Product availability
- Inventory constraints
- Shipping availability
- Operational capacity limitations

This step significantly reduces invalid decision variables before optimization.

3.4 Adaptive Priority-Based Candidate Reduction

Although feasibility filtering removes invalid assignments, the remaining candidate space is still large for direct quantum optimization. Therefore, an adaptive priority scoring mechanism is introduced.

The priority score evaluates each candidate assignment using multiple business factors:

- Revenue contribution
- Shipping cost
- Inventory availability
- Dock capacity utilization
- Throughput availability
- Penalty impact

The adaptive score is calculated to prioritize assignments that provide higher business value while maintaining operational feasibility.

After applying the candidate reduction strategy:

Stage	Number of Candidates
Initial feasible candidates	133,495
Reduced optimization candidates	65,857
Reduction achieved	50.67%

This reduction improves scalability and decreases the number of binary variables required for the QUBO formulation.

3.5 QUBO Formulation

The reduced DOM problem is converted into a Quadratic Unconstrained Binary Optimization (QUBO) model, which is suitable for quantum optimization algorithms.

Each candidate assignment is represented as a binary variable:

$$x_i \in \{0, 1\}$$

where:

- $x_i = 1$indicates that the assignment is selected.
- $x_i = 0$indicates that the assignment is not selected.

The QUBO objective function is represented as:

$$\min Q\,(x) = x^T Q x$$

where:

- $x$represents the binary decision vector.
- Qrepresents the QUBO coefficient matrix containing objective and penalty terms.

The QUBO formulation incorporates:

- Maximization of adaptive priority score
- Assignment uniqueness constraints
- Capacity limitations
- Business feasibility requirements

3.6 Hybrid Quantum-Classical Optimization

The formulated QUBO problem is solved using two optimization approaches:

Classical Exact QUBO Solver

The exact solver provides the optimal solution for smaller problem instances and acts as a benchmark for evaluating quantum results.

Quantum Approximate Optimization Algorithm (QAOA)

QAOA is applied to solve the QUBO problem using a quantum circuit-based optimization approach.

The QAOA process consists of:

1. Encoding the QUBO problem into an Ising Hamiltonian
2. Applying parameterized quantum circuits
3. Optimizing circuit parameters using a classical optimizer
4. Measuring the final quantum state to obtain candidate solutions

This hybrid process combines quantum exploration capability with classical optimization efficiency.

3.7 Business Repair and Validation Layer

The raw optimization output may contain solutions that require additional business-level validation. Therefore, a repair layer is introduced after optimization.

The repair layer performs:

- Duplicate order removal
- Constraint verification
- Assignment consistency checking
- Final feasibility validation

This ensures that the final solution satisfies real-world DOM requirements.

3.8 Final Solution Generation

The validated solution provides:

- Feasible order-to-distribution-center assignments
- Optimized revenue contribution
- Reduced shipping cost
- Minimum penalty impact

The final output is compared against classical baseline approaches to evaluate the effectiveness of the proposed hybrid quantum-classical framework.

4. Experimental Setup and Implementation Details

This section should be around **1.5 pages only**.

Include:

4.1 Computational Environment

Mention:

- Programming Language: Python
- Platform: Google Colab
- Quantum Framework: Qiskit
- Optimization Libraries:
  - Qiskit Optimization
  - NumPy
  - Pandas
  - SciPy

4.2 Dataset Description

Short table:

Component	Description
Orders	Customer order information
Inventory	Available stock at DCs
Shipping Cost	Transportation cost between locations
Dock Capacity	Loading limitations
Throughput	Processing capacity

4.3 Dataset Processing Results

Add your actual numbers:

Processing Stage	Count
Initial assignment candidates	352,702
Feasible assignments	133,495
Reduced candidates	65,857
QUBO variables	65,857

4.4 Optimization Configuration

Mention:

- QUBO variables represented as binary decision variables.
- QAOA used for quantum optimization.
- Classical optimizer: COBYLA.
- Benchmarking performed using:
    - Greedy baseline
    - Exact QUBO solver
    - QAOA solution

5. Experimental Results and Performance Analysis

5.1 Optimization Performance Evaluation

The proposed hybrid quantum-classical framework was evaluated using three optimization approaches:

1. Greedy Baseline Method
2. Exact QUBO Solver
3. Quantum Approximate Optimization Algorithm (QAOA)

The greedy method was used as a classical baseline, while the exact QUBO solver provided an optimal reference for the reduced problem size. QAOA was evaluated to analyze the capability of quantum optimization for solving the DOM assignment problem.

The performance was measured using the following evaluation metrics:

- Total revenue generated
- Total shipping cost
- Penalty incurred
- Adaptive priority score
- Number of feasible assignments

5.2 Comparison of Optimization Approaches

- The obtained results are summarized below:

Optimization Method	Final Assignments	Revenue	Shipping Cost	Penalty	Adaptive Score
Greedy Baseline	22,046	76,691,433	32,027,721	461.746	2,213.717
Exact QUBO Solver	15	4,744,085	22,405	0.0	6.662
QAOA Optimization	12	2,490,693	12,477	0.04	4.002

5.3 Analysis of Results

The greedy baseline method generated a large number of assignments and achieved high total revenue. However, it resulted in significantly higher shipping costs and penalties because it does not perform global optimization of assignment decisions.

The Exact QUBO solver produced the highest-quality solution among the optimization approaches for the reduced problem. It achieved:

- Zero penalty violations
- Lower transportation cost
- Higher adaptive optimization score

This demonstrates the effectiveness of the QUBO formulation in representing DOM constraints.

The QAOA-based quantum optimization approach successfully generated feasible solutions after applying the business repair layer. Although the solution quality was lower compared with the exact solver, it demonstrated the ability of quantum optimization methods to solve constrained supply chain optimization problems.

5.4 Search Space Reduction Analysis The candidate reduction strategy significantly decreased the optimization complexity before applying quantum algorithms.

Processing Stage	Number of Candidates
Initial assignment candidates	352,702
Feasible assignments	133,495
Reduced QUBO candidates	65,857

The adaptive reduction mechanism reduced the QUBO search space by:

$$50.67\%$$

This reduction is important because current quantum devices have limited qubit availability. By reducing unnecessary candidate assignments, the proposed approach improves the scalability of quantum optimization.

5.5 Business Repair Layer Evaluation

A business repair layer was introduced to convert raw optimization outputs into practical DOM decisions.

The repair process included:

- Removing duplicate order selections
- Checking assignment validity
- Ensuring business constraints are satisfied

The quantum output before repair contained multiple selected candidates for certain orders due to the reduced QUBO representation. After applying the repair mechanism, the solution was converted into a valid business assignment.

Results:

Stage	Selected Assignments
Raw Quantum Output	15
After Business Repair	12

The repair layer improves the practical usability of quantum-generated solutions by ensuring compatibility with real-world supply chain requirements.

6. Conclusion and Future Work

6.1 Conclusion

This project presented a **hybrid quantum-classical optimization framework** for solving the Distributed Order Management (DOM) problem using the WISER challenge dataset. The proposed approach combines classical data preprocessing and candidate reduction techniques with quantum optimization methods to efficiently solve large-scale order assignment problems.

The framework began by integrating multiple operational datasets, including orders, inventory, shipping costs, dock capacity, and throughput information. A feasibility analysis generated valid order-to-distribution-center assignments, while an adaptive priority scoring mechanism evaluated each candidate based on revenue, shipping cost, inventory availability, operational capacity, and penalty factors. This adaptive filtering significantly reduced the optimization search space, making the problem suitable for quantum optimization.

The reduced optimization problem was formulated as a **Quadratic Unconstrained Binary Optimization (QUBO)** model and solved using both an Exact QUBO Solver and the **Quantum Approximate Optimization Algorithm (QAOA).** A business repair layer was introduced to validate the optimization results, eliminate duplicate assignments, and ensure that the final solution satisfied practical business constraints.

The experimental results demonstrated that the proposed framework successfully reduced the optimization complexity while producing feasible business solutions. The comparison between the greedy baseline, Exact QUBO Solver, and QAOA highlighted the potential of hybrid quantum-classical optimization for solving combinatorial supply chain problems. Although current quantum hardware limits the size of problems that can be executed directly, the proposed methodology establishes a scalable workflow that can benefit from future advances in quantum computing.

Overall, this work demonstrates that integrating adaptive preprocessing, QUBO modeling, quantum optimization, and business validation provides an effective approach for tackling complex Distributed Order Management problems.

6.2 Future Work

The proposed framework can be extended in several directions to further improve its performance and applicability.

- Evaluate the framework on larger real-world supply chain datasets using next-generation quantum hardware.
- Investigate advanced quantum optimization algorithms with improved parameter optimization and circuit design.
- Develop multi-objective optimization models that simultaneously optimize cost, service level, sustainability, and delivery time.
- Incorporate real-time inventory updates and dynamic customer demand to support continuous decision-making.
- Explore hybrid decomposition techniques that partition very large DOM problems into multiple quantum-solvable subproblems.
- Compare the proposed framework with additional classical optimization approaches such as Mixed Integer Linear Programming (MILP), Genetic Algorithms, and Simulated Annealing.
- Integrate explainable decision-support mechanisms to improve the interpretability of optimization results for supply chain managers.

The continued advancement of quantum computing technologies is expected to enable the solution of increasingly large and complex optimization problems. The proposed hybrid framework provides a strong foundation for future research in quantum-assisted supply chain optimization and intelligent Distributed Order Management systems.